

# Material & Landscape Palette

131 trees across 5 desert species, and the materials that carry the crescent's language.

09

OF 12 UPLOAD SLOTS

131

TREES PLANTED

5

DESERT SPECIES

AED 1,798

PER M<sup>2</sup> OF SITE

2,595 m<sup>2</sup>

CANOPY AREA

▶ VISIT THE LIVE PROJECT PORTAL

[wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

01

## WHAT IS IN THIS FILE

Phase6 Detailed Design Report

DRAWING

Planting

DRAWING

Section

DRAWING

02

## THE SCALE OF THE ANALYSIS BEHIND IT

39 years

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours

of sun position ray-traced against 131 trees for every hour of the year

15,000

one-metre ground cells modelled for shade and comfort, site-wide

4 models

trained and tested for leakage — two of them decide what this document reports

41 checks

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

03

## THE PHASES BEHIND THIS FILE

P6 Detailed Design

### VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository [github.com/wasim437/AI\\_SAFA](https://github.com/wasim437/AI_SAFA)

Live portal [wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

Produced by [CODE](#) [src/config.py](#) [src/plan.py](#) [src/drawings.py](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

# A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

09

OF 12

01

## THE TEN PHASES

### P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

### P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

### P5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m<sup>2</sup>

### P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

### P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

### P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

### P4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

### P6 Detailed Design

THIS DOCUMENT

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

### P8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

### P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02

## THE CODE AND DATA BEHIND THIS DOCUMENT

Config

src

Plan

src

Drawings

src

03

## REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 04 OF 12 · KEY SECTIONS, ELEVATIONS &amp; MATERIAL PALETTE

# Key Sections, Elevations & Material Palette

The canopy section solved against the shadow geometry

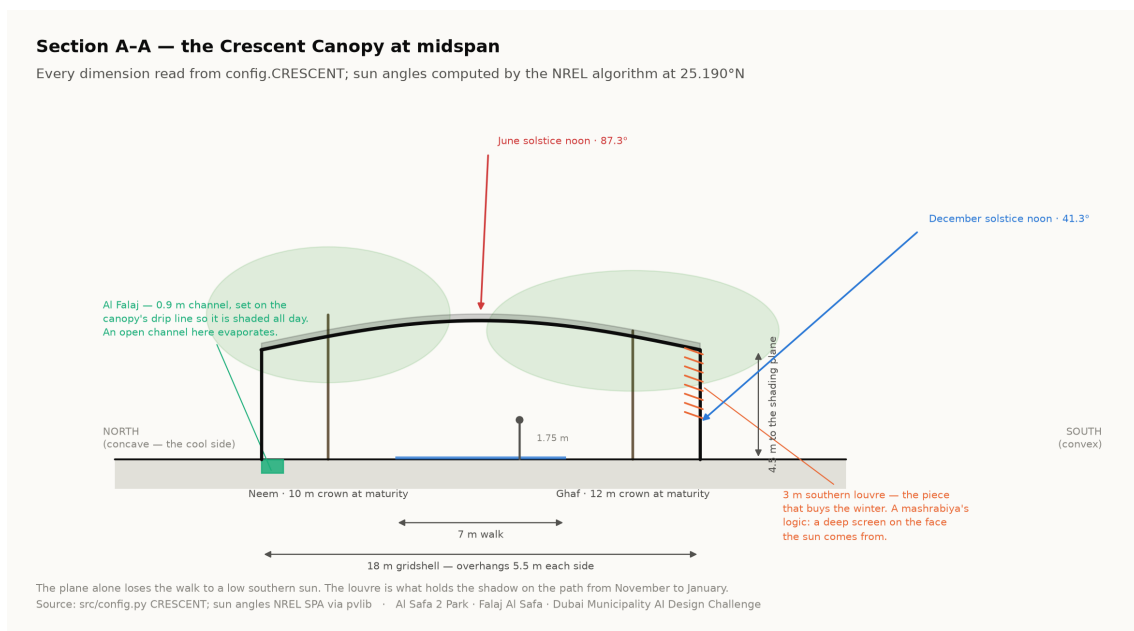
This is the drawing the whole scheme rests on: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m louvre on the southern face. Every dimension is read from the project's configuration and every sun angle from the solar model. None of it is drawn by eye.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m<sup>2</sup> · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by python `run_analysis.py`

## 1. The section, and the problem it solves

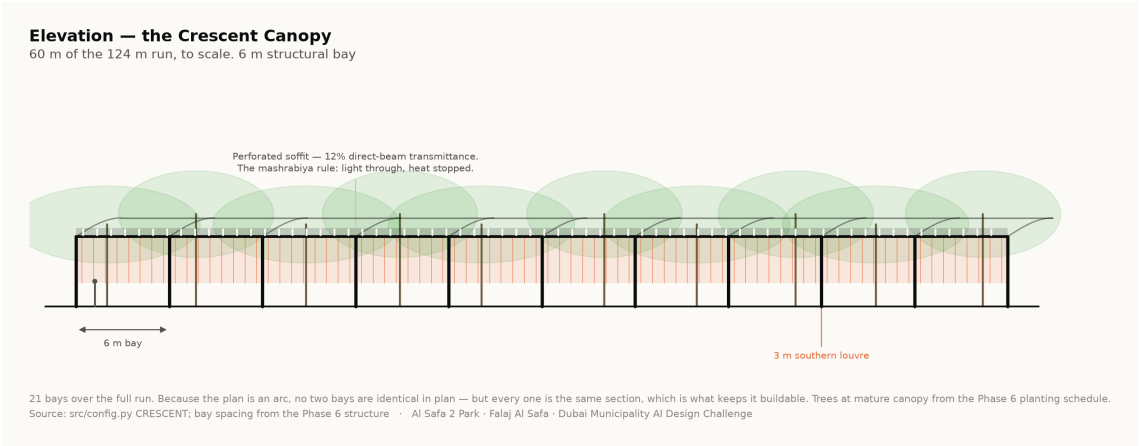
A flat canopy fails in a specific and predictable way. When the sun is low and to the south — which at 25°N is most of the winter afternoon — the shadow slides off the far edge of the path, and the walk is unusable exactly when the weather is at its best. An earlier version of this project claimed 99.2% annual shade on such a canopy. It was withdrawn.

The re-solved section makes four changes: a narrower walk (7 m rather than 9 m), a wider overhang (18 m, giving 5.5 m each side), a lower plane (4.5 m), and a 3 m vertical louvre on the southern face. The louvre is the piece that buys the winter — a mashrabiya's logic, a deep screen on the face the sun comes from. The section now measures 87.3%.



Section A-A at midspan. Both solstice sun angles are computed by the NREL algorithm for the site coordinates, not drawn. Technical drawing — to scale.

## 2. Elevation and structural rhythm



Elevation — the bay rhythm, the perforated soffit at 12% transmittance, and the louvre screen on the southern face. Technical drawing — to scale.

The gridshell spans 18 m on regular structural bays. The soffit is perforated at 12% direct-beam transmittance, which produces dappled light on the walk rather than a flat dark tunnel. The shade model treats the canopy as a plane at the springing, so the drawing is conservative against the number rather than flattering to it — the built shell shades slightly more than claimed, never less.

3. Why the tree avenue exists

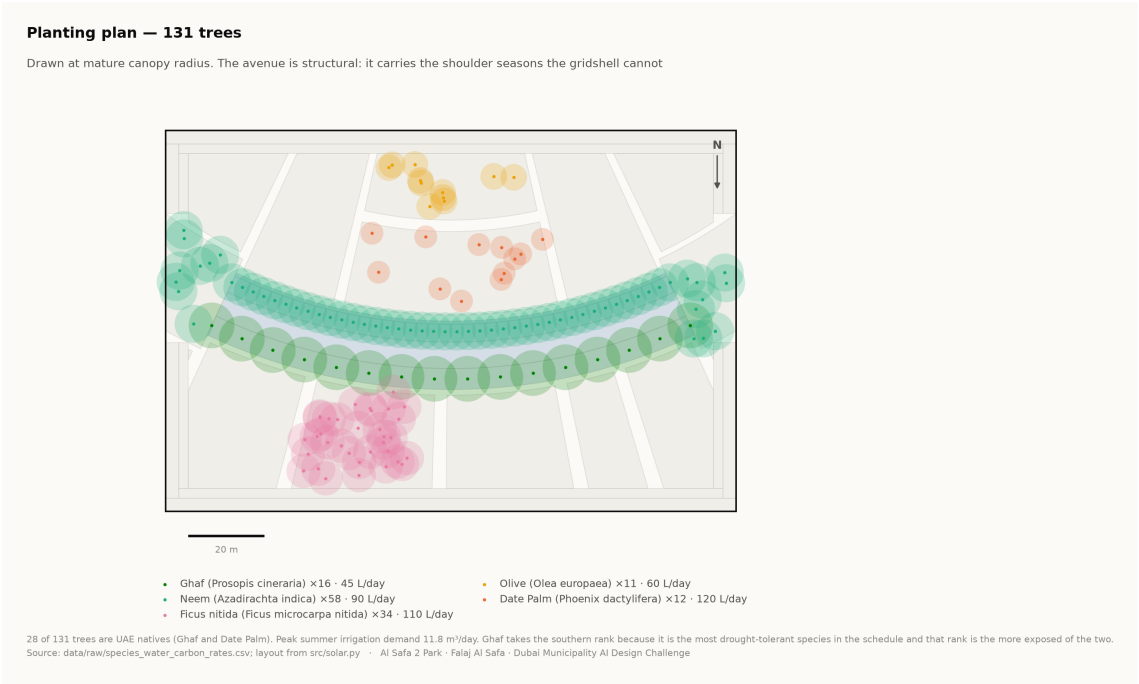
Shadow length is height divided by the tangent of solar elevation. At Dubai's summer noon a 6 m tree casts a 0.19 m shadow; at 20° elevation the same tree casts 16.5 m. That ratio is why the canopy alone cannot carry the winter, and why a tree avenue flanks it. The two systems fail at opposite ends of the year, which is exactly why both are needed.

4. Material and landscape palette

- **Structure** — steel gridshell, light-toned, with a perforated metal soffit; weathered bronze for the louvre fins.
- **Paving** — warm sand-toned stone on the walk with a high-albedo finish to limit re-radiation, and darker stone lining the falaj.
- **Earth** — Al Kathib is planted earth, not a wall. Cut and fill are balanced on site where possible.
- **Planting** — five desert species only. Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.

| Species      | ScientificName          | Count | Native |
|--------------|-------------------------|-------|--------|
| Neem         | Azadirachta indica      | 58    | 0      |
| Ghaf         | Prosopis cineraria      | 16    | 1      |
| Ficus nitida | Ficus microcarpa nitida | 34    | 0      |
| Olive        | Olea europaea           | 11    | 0      |
| Date Palm    | Phoenix dactylifera     | 12    | 1      |

131 trees. Source: data/raw/species\_water\_carbon\_rates.csv.



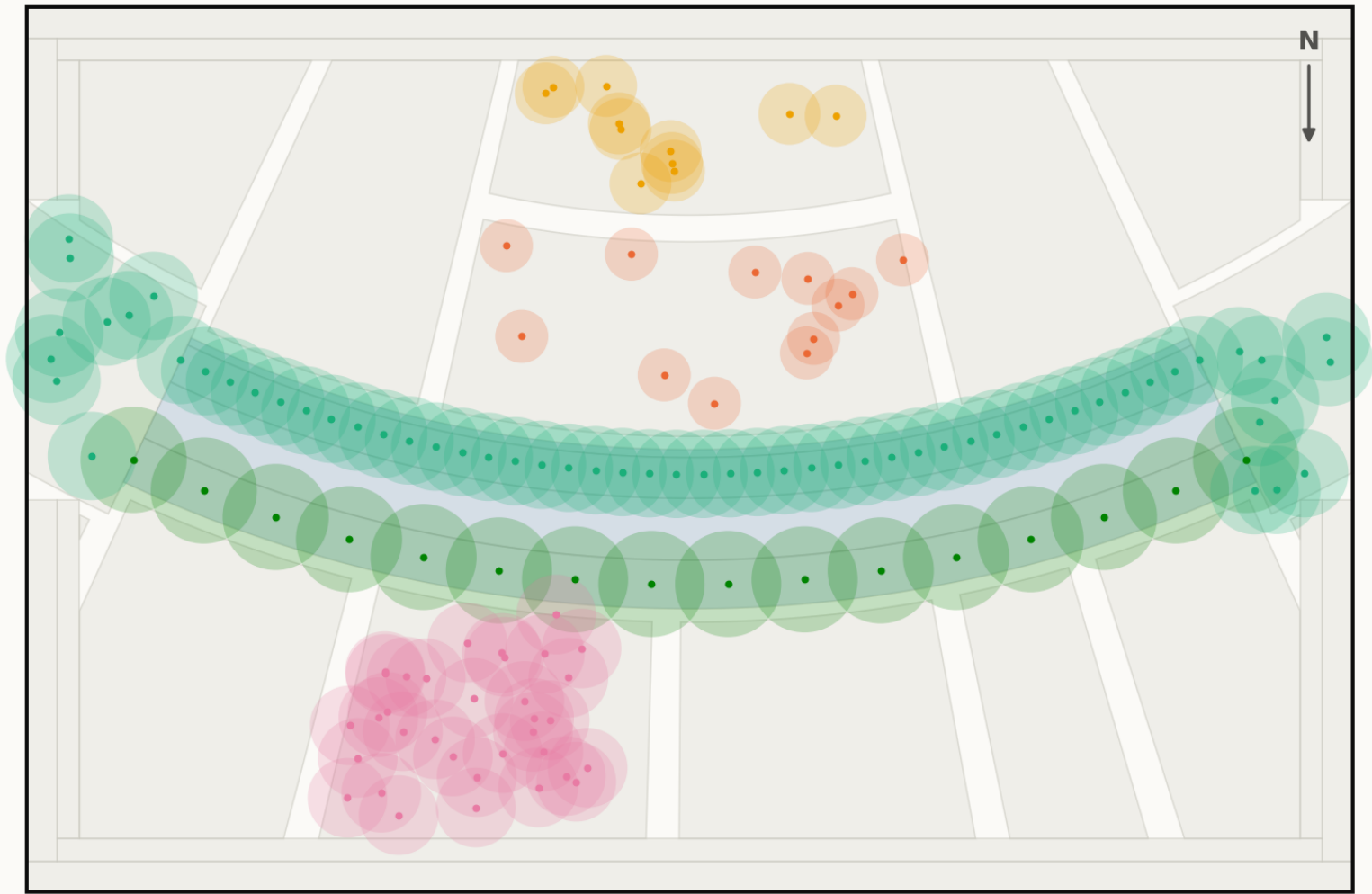
Planting plan — 131 trees drawn at mature canopy radius. Technical drawing — to scale.

Planting

Technical drawing — to scale.

Planting plan — 131 trees

Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot

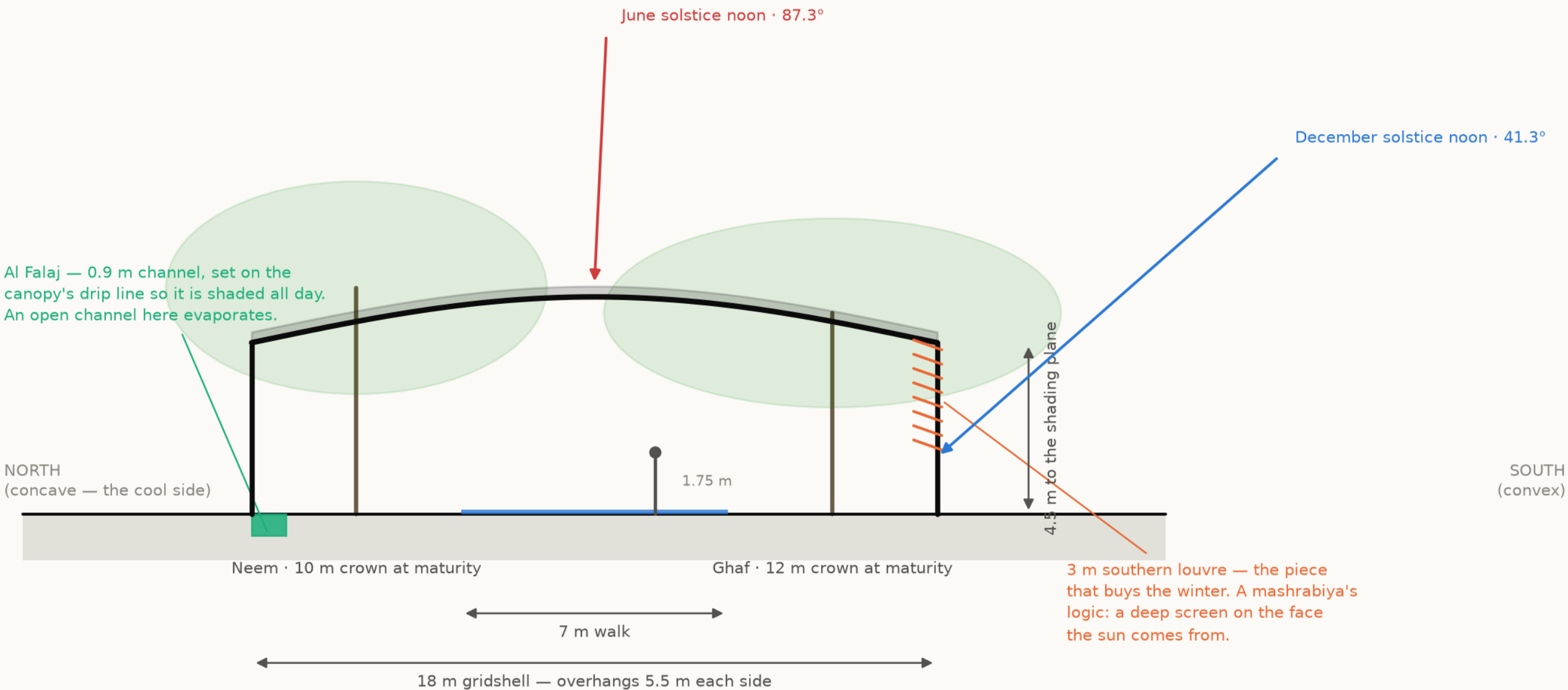


- Ghaf (*Prosopis cineraria*) ×16 · 45 L/day
- Olive (*Olea europaea*) ×11 · 60 L/day
- Neem (*Azadirachta indica*) ×58 · 90 L/day
- Date Palm (*Phoenix dactylifera*) ×12 · 120 L/day
- Ficus nitida (*Ficus microcarpa nitida*) ×34 · 110 L/day

28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.  
Source: data/raw/species\_water\_carbon\_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



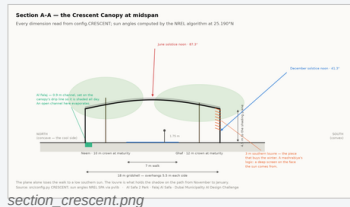
The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.  
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

# Every step of the work, with its proof attached

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out.  
Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

## PHASE 6 Detailed Design

THIS DOCUMENT RESTS ON



section\_crescent.png

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

CODE [src/drawings.py](#) [src/config.py](#)

PROOF — click to open [data/processed/planting\\_layout.csv](#)

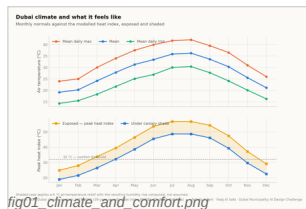


fig01\_climate\_and\_comfort.png

## PHASE 1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

CODE [src/climate.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/hourly\\_climate\\_comfort\\_8760.csv](#) [data/raw/sources.json](#)

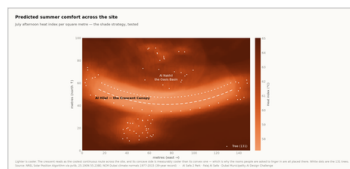


fig04\_site\_comfort\_map.png

## PHASE 2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

CODE [src/climate.py](#)

PROOF — click to open [models/headline\\_metrics.json](#)

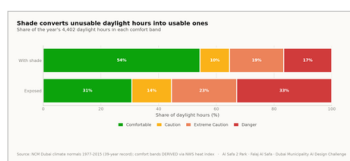


fig02\_comfort\_bands.png

## PHASE 3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

CODE [src/config.py](#) [src/plan.py](#)

PROOF — click to open [models/headline\\_metrics.json](#)

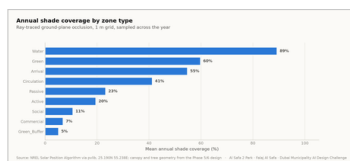


fig03\_shade\_by\_zone.png

## PHASE 4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

CODE [src/plan.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/masterplan\\_geometry.json](#)



# The work, with its proof attached — continued

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.



fig10\_masterplan.png

## PHASE 5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

CODE [src/plan.py](#) [src/figures.py](#)

PROOF — [click to open data/raw/site\\_zoning\\_schedule.csv](#)  
[data/processed/masterplan\\_geometry.json](#)

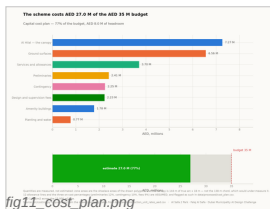


fig11\_cost\_plan.png

## PHASE 7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

CODE [src/costing.py](#) [src/climate.py](#)

PROOF — [click to open data/processed/cost\\_plan.csv](#) [models/cost\\_summary.json](#)

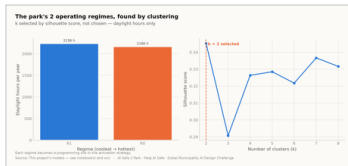


fig08\_microclimate\_regimes.png

## PHASE 8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

CODE [src/models.py](#) [src/dataset.py](#)

PROOF — [click to open data/processed/spatial\\_grid\\_comfort.csv](#)

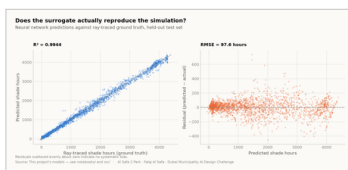


fig05\_surrogate\_performance.png

## PHASE 9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

CODE [src/models.py](#) [src/boards.py](#) [tools/sync\\_film.py](#)

PROOF — [click to open models/model\\_metrics.json](#) [tests/test\\_pipeline.py](#)



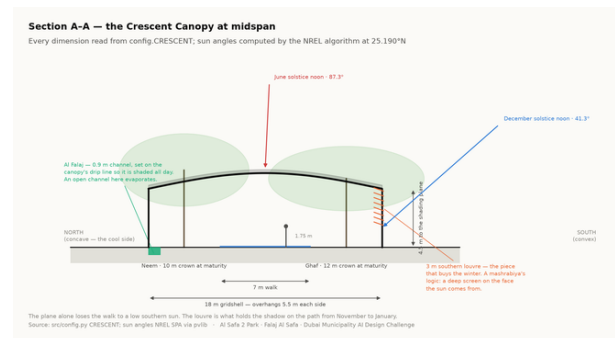
## PHASE 10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

CODE [tools/build\\_submission\\_pdfs.py](#) [tools/build\\_reports.py](#)

PROOF — [click to open tests/test\\_pipeline.py](#)

## 01 IN THIS FILE



## Section

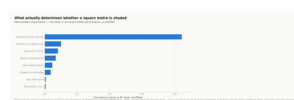
*technical drawing*

**Shade converts unusable daylight hours into usable ones**  
 (Share of the year's 6,922 daylight hours in each comfort band)

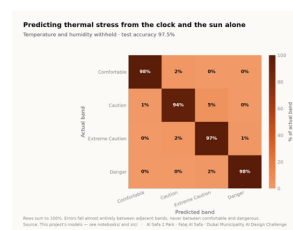
| Scenario      | Comfortable | Caution | Extreme Caution | Danger |
|---------------|-------------|---------|-----------------|--------|
| Without shade | 34%         | 33%     | 20%             | 13%    |
| With shade    | 53%         | 15%     | 23%             | 9%     |

Source: WSP Solar climate research (2017-2020 10-year average), comfort bands 2019/20 based on 2019/20 values. At: 04/01/2021, 10:02:10 AM. Source: Knowledge @ Orange World.

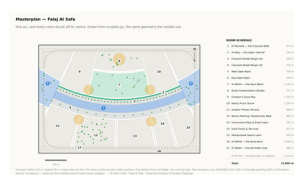
**Fig02 comfort bands**  
*analysis output*



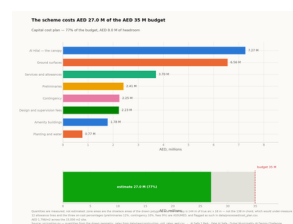
**Fig04 site comfort map**  
*analysis output*



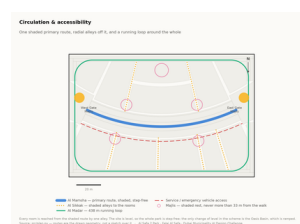
**Fig06 feature importance**  
*analysis output*



**Fig10 masterplan**  
*analysis output*



**Fig11 cost plan**  
*analysis output*



## Circulation

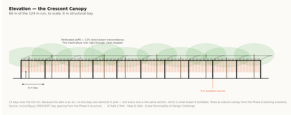
*technical drawing*

# The visual record — continued

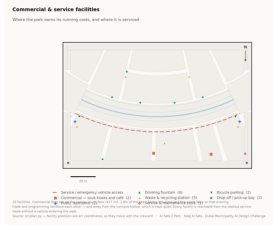
This slot's own pictures come first, larger. The remaining 16 of the 18 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

09

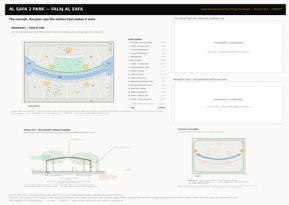
OF 12



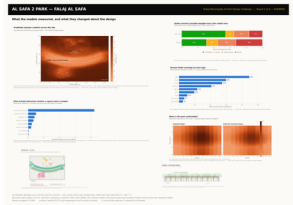
**Elevation**  
technical drawing



**Facilities**  
technical drawing



**Board 1 concept**  
presentation board



**Board 2 evidence**  
presentation board

# Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

09

OF 12

## THE FIGURES THIS FILE LEADS ON

each one appears in the full tables below, from the same file

131

TREES PLANTED

5

DESERT SPECIES

AED 1,798

PER M<sup>2</sup> OF SITE

2,595 m<sup>2</sup>

CANOPY AREA

### MEASURED RESULT

models/headline\_metrics.json

|                                          |         |
|------------------------------------------|---------|
| Annual daylight hours modelled           | 4,402   |
| Comfortable daylight hours — today       | 44.5%   |
| Comfortable daylight hours — as designed | 64.6%   |
| Mean heat-index reduction under canopy   | 7.13 °C |
| Peak heat index, exposed                 | 56.8 °C |
| Peak heat index, shaded                  | 48.7 °C |
| Crescent Walk shaded, canopy and louvre  | 87.3%   |
| Site-wide mean shade                     | 34.1%   |
| Trees planted                            | 131     |

### TRAINED MODELS, ON HELD-OUT TEST SETS

models/model\_metrics.json

|                                                              |                        |
|--------------------------------------------------------------|------------------------|
| M1a Random Forest — shade surrogate, test R <sup>2</sup>     | 0.9982                 |
| M1b Neural network — deployed surrogate, test R <sup>2</sup> | 0.9944                 |
| M2 Gradient Boosting — comfort band, test accuracy           | 97.5%                  |
| M3 K-Means — regimes selected by silhouette                  | k = 2                  |
| Training / validation / test split                           | 10,500 / 2,250 / 2,250 |
| Random seed, fixed for reproducibility                       | 42                     |

### WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

#### Climate fidelity

the modelled year reproduces the published normals it was built from

#### No overlaps

no room in the schedule overlaps another

#### Areas close

every area sums back to 15,000 m<sup>2</sup>

#### Budget held

the cost plan stays inside AED 35 M

#### No leakage

no model can see its own answer in its inputs

### RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js   # 64 portal checks
node tests/test_film.js          # every frame of the film
```

# The complete project, and where to check it

Every one of the 120 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository [github.com/wasim437/AI\\_SAFA](https://github.com/wasim437/AI_SAFA)  
Live portal [wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

## THE SOURCES BEHIND THIS FILE

*open these first if you want to check this document*

|                                 |                               |
|---------------------------------|-------------------------------|
| <a href="#">src/config.py</a>   | Technical drawing — to scale. |
| <a href="#">src/plan.py</a>     | Technical drawing — to scale. |
| <a href="#">src/drawings.py</a> | Technical drawing — to scale. |

## THE TWELVE UPLOAD FILES (12)

|                                                                                              |                                              |
|----------------------------------------------------------------------------------------------|----------------------------------------------|
| <a href="#">UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf</a>                    | Design Narrative & Concept                   |
| <a href="#">UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf</a> | Neighborhood Park Preliminary Design Ma...   |
| <a href="#">UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf</a> | Concept Plans and Spatial Organization Di... |
| <a href="#">UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf</a>                     | Key Sections & Elevations                    |
| <a href="#">UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf</a>                   | 3D & Spatial Visualizations                  |
| <a href="#">UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf</a>                           | AI Methodology Report                        |
| <a href="#">UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf</a>         | User Experience & Activation Strategy        |
| <a href="#">UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf</a>             | Sustainability Concept & Strategy            |
| <a href="#">UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf</a>                  | Material & Landscape Palette                 |
| <a href="#">UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf</a>                          | Complete Design Report                       |
| <a href="#">UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf</a>        | Site Analysis & Human-Centric Research       |
| <a href="#">UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf</a>                    | One-minute Concept Animation                 |

## PROJECT DOCUMENTS (7)

|                                                   |                                                |
|---------------------------------------------------|------------------------------------------------|
| <a href="#">AL_SAFA_MASTER_PROMPT.md</a>          | The whole design, stated for visualisation     |
| <a href="#">DATA_SOURCES.md</a>                   | Every source, its period, and its limitations  |
| <a href="#">LINKS.md</a>                          | Every public URL this submission points at     |
| <a href="#">PROJECT_PLAN.md</a>                   | Requirements, phases, status, and what is left |
| <a href="#">PUSH_TO_GITHUB.md</a>                 | Project document                               |
| <a href="#">README.md</a>                         | The design argument, written for a juror       |
| <a href="#">EXPLAIN_THE_PROJECT/START_HERE.md</a> | The project in plain language                  |

## THE WRITTEN REPORTS (11)

|                                                                              |                                  |
|------------------------------------------------------------------------------|----------------------------------|
| <a href="#">reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf</a>        | Generated from the live analysis |
| <a href="#">reports/pdf/Concept_Animation_Storyboard.pdf</a>                 | Generated from the live analysis |
| <a href="#">reports/pdf/Phase2_Problem_Definition_Report.pdf</a>             | Generated from the live analysis |
| <a href="#">reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf</a>     | Generated from the live analysis |
| <a href="#">reports/pdf/Phase4_Concept_Development_Report.pdf</a>            | Generated from the live analysis |
| <a href="#">reports/pdf/Phase5_Masterplan_Development_Report.pdf</a>         | Generated from the live analysis |
| <a href="#">reports/pdf/Phase6_Detailed_Design_Report.pdf</a>                | Generated from the live analysis |
| <a href="#">reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf</a> | Generated from the live analysis |
| <a href="#">reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf</a> | Generated from the live analysis |
| <a href="#">reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf</a>  | Generated from the live analysis |
| <a href="#">reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf</a>  | Generated from the live analysis |

## THE ANALYSIS FIGURES (11)

|                                                       |                                              |
|-------------------------------------------------------|----------------------------------------------|
| <a href="#">figures/fig01_climate_and_comfort.png</a> | Analysis output — computed from project data |
| <a href="#">figures/fig02_comfort_bands.png</a>       | Analysis output — computed from project data |
| <a href="#">figures/fig03_shade_by_zone.png</a>       | Analysis output — computed from project data |

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[figures/fig04\\_site\\_comfort\\_map.png](#)  
[figures/fig05\\_surrogate\\_performance.png](#)  
[figures/fig06\\_feature\\_importance.png](#)  
[figures/fig07\\_confusion\\_matrix.png](#)  
[figures/fig08\\_microclimate\\_regimes.png](#)  
[figures/fig09\\_diurnal\\_comfort.png](#)  
[figures/fig10\\_masterplan.png](#)  
[figures/fig11\\_cost\\_plan.png](#)

Analysis output — computed from project data  
Analysis output — computed from project data  
Analysis output — computed from project data  
Analysis output — computed from project data  
Analysis output — computed from project data  
Analysis output — computed from project data  
Analysis output — computed from project data  
Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation\\_crescent.png](#)  
[design/visuals/elevation\\_crescent.png](#)  
[design/visuals/facilities\\_crescent.png](#)  
[design/visuals/planting\\_crescent.png](#)  
[design/visuals/section\\_crescent.png](#)  
[design/boards/board\\_1\\_concept.png](#)  
[design/boards/board\\_2\\_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)  
Technical drawing — generated from [src/plan.py](#)  
Technical drawing — generated from [src/plan.py](#)  
Technical drawing — generated from [src/plan.py](#)  
Technical drawing — generated from [src/plan.py](#)  
Technical drawing — generated from [src/plan.py](#)  
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/\\_\\_init\\_\\_.py](#)  
[src/boards.py](#)  
[src/climate.py](#)  
[src/config.py](#)  
[src/costing.py](#)  
[src/dataset.py](#)  
[src/drawings.py](#)  
[src/figures.py](#)  
[src/models.py](#)  
[src/plan.py](#)  
[src/solar.py](#)  
[src/viz.py](#)  
[run\\_analysis.py](#)

Analysis module  
The two presentation boards  
The 8,760-hour year rebuilt from NCM normals  
Every constant the project reads  
The cost model against the AED 35 M ceiling  
Assembles the ML training tables  
Section, elevation, circulation, planting, facilities  
The analysis charts  
The four models  
SINGLE SOURCE of the crescent geometry  
Sun position and shadow ray-tracing  
Analysis module  
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

[tools/\\_\\_init\\_\\_.py](#)  
[tools/build\\_docs.py](#)  
[tools/build\\_links.py](#)  
[tools/build\\_notebook.py](#)  
[tools/build\\_reports.py](#)  
[tools/build\\_submission\\_pdfs.py](#)  
[tools/organize\\_repo.py](#)  
[tools/report\\_content.py](#)  
[tools/sync\\_film.py](#)  
[tools/sync\\_portal.py](#)  
[tools/sync\\_submission.py](#)

THE DATA, AS ISSUED (7)

[data/raw/construction\\_unit\\_rates\\_aed.csv](#)  
[data/raw/dubai\\_climate\\_normals\\_ncm.csv](#)  
[data/raw/dubai\\_population\\_dsc\\_2023.csv](#)  
[data/raw/site\\_zoning\\_schedule.csv](#)  
[data/raw/sources.json](#)  
[data/raw/species\\_water\\_carbon\\_rates.csv](#)  
[data/raw/walk\\_catchment\\_rings.csv](#)

Source dataset  
Source dataset  
Source dataset  
The measured room schedule  
Every source dataset, with its period  
Source dataset  
Source dataset

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# The complete project — continued

## THE DATA, AS PROCESSED (6)

|                                                                |                                           |
|----------------------------------------------------------------|-------------------------------------------|
| <a href="#">data/processed/cost_plan.csv</a>                   | The capital cost plan, line by line       |
| <a href="#">data/processed/hourly_climate_comfort_8760.csv</a> | The 8,760-hour climate and comfort series |
| <a href="#">data/processed/masterplan_geometry.json</a>        | The plan geometry, as exported            |
| <a href="#">data/processed/planting_layout.csv</a>             | Every one of the 131 trees                |
| <a href="#">data/processed/schema.json</a>                     | Generated by the pipeline                 |
| <a href="#">data/processed/spatial_grid_comfort.csv</a>        | The 15,000-cell ground grid               |

## THE TRAINED MODELS (3)

|                                              |                       |
|----------------------------------------------|-----------------------|
| <a href="#">models/cost_summary.json</a>     | Model output          |
| <a href="#">models/headline_metrics.json</a> | The headline numbers  |
| <a href="#">models/model_metrics.json</a>    | Trained-model metrics |

## THE TESTS (2)

|                                        |                                 |
|----------------------------------------|---------------------------------|
| <a href="#">tests/test_pipeline.py</a> | 41 correctness checks           |
| <a href="#">tests/test_film.js</a>     | Every frame of the concept film |

## THE NOTEBOOK (1)

|                                                                  |                                         |
|------------------------------------------------------------------|-----------------------------------------|
| <a href="#">notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb</a> | The complete analysis, outputs embedded |
|------------------------------------------------------------------|-----------------------------------------|

## THE WEBSITE AND ANALYTICS PORTAL (9)

|                                               |                     |
|-----------------------------------------------|---------------------|
| <a href="#">docs/index.html</a>               | The project website |
| <a href="#">docs/SUBMISSION_GUIDE.md</a>      | Published site      |
| <a href="#">docs/_PORTAL/gallery_data.js</a>  | Published site      |
| <a href="#">docs/_PORTAL/plan_geometry.js</a> | Published site      |
| <a href="#">docs/_PORTAL/portal.js</a>        | Published site      |
| <a href="#">docs/_PORTAL/portal_data.js</a>   | Published site      |
| <a href="#">docs/_PORTAL/selftest.js</a>      | Published site      |
| <a href="#">docs/_PORTAL/DATA_AUDIT.md</a>    | Published site      |
| <a href="#">docs/_PORTAL/README.md</a>        | Published site      |

## THE COMPETITION BRIEF, AS ISSUED (7)

|                                                                                             |                                    |
|---------------------------------------------------------------------------------------------|------------------------------------|
| <a href="#">00_BRIEF/Ai Park - Master Plan (4).jpg</a>                                      | Dubai Municipality source document |
| <a href="#">00_BRIEF/Ai Park Scope of Work &amp; General Terms &amp; Conditions (5).pdf</a> | Dubai Municipality source document |
| <a href="#">00_BRIEF/Ai Safa Park 2 Plan (5).dwg</a>                                        | Dubai Municipality source document |
| <a href="#">00_BRIEF/MAIN WEBSITE DETAILS.txt</a>                                           | Dubai Municipality source document |
| <a href="#">00_BRIEF/Neighborhood Parks Manual (4).pdf</a>                                  | Dubai Municipality source document |
| <a href="#">00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt</a>                                 | Dubai Municipality source document |
| <a href="#">00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt</a>                                        | Dubai Municipality source document |

## THE TWELVE SUBMISSION FOLDERS (12)

|                                                                                |                                                  |
|--------------------------------------------------------------------------------|--------------------------------------------------|
| <a href="#">submission/01_Design_Narrative_Concept/MANIFEST.md</a>             | What is in the slot, and what produced each file |
| <a href="#">submission/02_Preliminary_Design_Masterplan/MANIFEST.md</a>        | What is in the slot, and what produced each file |
| <a href="#">submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md</a>       | What is in the slot, and what produced each file |
| <a href="#">submission/04_Key_Sections_Elevations/MANIFEST.md</a>              | What is in the slot, and what produced each file |
| <a href="#">submission/05_3D_Spatial_Visualizations/MANIFEST.md</a>            | What is in the slot, and what produced each file |
| <a href="#">submission/06_AI_Methodology_Report/MANIFEST.md</a>                | What is in the slot, and what produced each file |
| <a href="#">submission/07_User_Experience_Activation_Strategy/MANIFEST.md</a>  | What is in the slot, and what produced each file |
| <a href="#">submission/08_Sustainability_Concept_Strategy/MANIFEST.md</a>      | What is in the slot, and what produced each file |
| <a href="#">submission/09_Material_Landscape_Palette/MANIFEST.md</a>           | What is in the slot, and what produced each file |
| <a href="#">submission/10_Complete_Design_Report/MANIFEST.md</a>               | What is in the slot, and what produced each file |
| <a href="#">submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md</a> | What is in the slot, and what produced each file |
| <a href="#">submission/12_Concept_Animation_Video/MANIFEST.md</a>              | What is in the slot, and what produced each file |

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# The complete project — continued

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WORKING HISTORY (1)

[archive/withdrawn\\_visuals/README.md](#)

Images withdrawn on purpose, and why

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